

WIPRO CAPSTONE PROJECT REPORT

Name	Hariharan M
Superset ID	5188144
Organization	Wipro
Project Type	Capstone Project
Project Name	Automation Testing using Selenium PyTest Framework
	Automation Testing using Behave BDD Framework
Domain	Travel and Booking
Module Title	Ixigo Trip Planner

SYNOPSIS

The Wipro Capstone Project focuses on automating the ixigo Trip Planner module using two separate automation frameworks: Selenium WebDriver with Python PyTest framework and Behave BDD framework. The automation validates travel month selection, origin city entry, filter selection, tourist location navigation, map verification, and flight booking workflow.

The framework was designed using the Page Object Model (POM) architecture to improve maintainability and reusability. Screenshots, logs, reports, and reusable utilities were integrated to support execution tracking and debugging.

The project includes both positive and negative testing scenarios using industry-standard automation testing practices.

PROJECT DESCRIPTION

The automation framework automates the end-to-end workflow of the ixigo Trip Planner module. The framework performs destination planning, filtering, location selection, Google Maps verification, and flight detail validation.

Separate page classes were implemented for Login, Planner, Filter, Location, Map, and Flight modules using Selenium Python Page Object Model framework.

The project also includes Behave BDD implementation where business scenarios are written in Gherkin syntax and integrated with Selenium automation scripts.

SELENIUM PYTEST FRAMEWORK

TEST STRATEGY

The project uses Selenium WebDriver with Python PyTest framework for automation testing. The framework supports:

- End-to-End Testing
- Positive Testing
- Negative Testing
- Data-Driven Testing

Tools Used:

- Selenium WebDriver
- PyTest
- Allure Reports
- Logging Module

TEST ENVIRONMENT

Operating System	Windows 10
IDE	PyCharm Unified Product
Browser	Google Chrome (or) Edge
Framework	Selenium WebDriver + PyTest
Design Pattern	Page Object Model (POM)
Reporting Tool	Allure Reports

FRAMEWORK ARCHITECTURE

IxigoTripPlanner

```
├─ config
├─ logs
├─ pages
├─ reports
│   ├── allure-results
│   └─ allure-report
│
├─ screenshots
├─ testdata
├─ tests
├─ utils
├─ conftest.py
├─ pytest.ini
└─ requirements.txt
```

TEST EXECUTION

The automation suite includes:

- 1 End-to-End Test Case
- 4 Positive Test Cases
- 2 Negative Test Cases

The End-to-End test validates the complete ixigo Trip Planner workflow from planner navigation until flight details verification.

SELENIUM TEST CASE SUMMARY

S.No	Test Type	Test Scenario	Expected Result	Status
1	End-to-End	Complete Trip Planner Workflow	User successfully reaches flight details page	Passed
2	Positive	Plan Page Navigation	Planner page opens successfully	Passed
3	Positive	Travel Month Selection	Selected month displayed successfully	Passed
4	Positive	Filter Selection Validation	Filter applied successfully	Passed
5	Positive	Tourist Location Selection	Tourist location page opens successfully	Passed
6	Negative	Invalid Location Search	Error message displayed successfully	Passed
7	Negative	Book Now Button Validation	Book Now button hidden for invalid route	Passed

END-TO-END FLOW VALIDATION

The End-to-End automation validates the complete business workflow including:

- User Login
- Planner Navigation
- Travel Month Selection
- Source Location Selection
- Travel Filter Application
- Tourist Location Selection
- Google Map Verification
- Book Now Navigation
- Flight Details Verification

POSITIVE TEST CASES

Positive Test Case 1 - Plan Page Navigation

- Validates successful navigation to the ixigo Trip Planner page
- Verifies Plan module loading
- Checks planner URL navigation

Positive Test Case 2 - Travel Month Selection

- Validates travel month selection functionality
- Reads month dynamically from Excel
- Verifies selected month update on the planner page

Positive Test Case 3 - Filter Selection Validation

- Validates category filter functionality
- Selects "International" category
- Selects "Pollution Free" filter
- Verifies filter application successfully

Positive Test Case 4 - Location Card Selection Validation

- Validates tourist location card selection
- Selects destination country
- Opens selected tourist location card
- Verifies successful location navigation

NEGATIVE TEST CASES

Negative Test Case 1 - Invalid Location Search

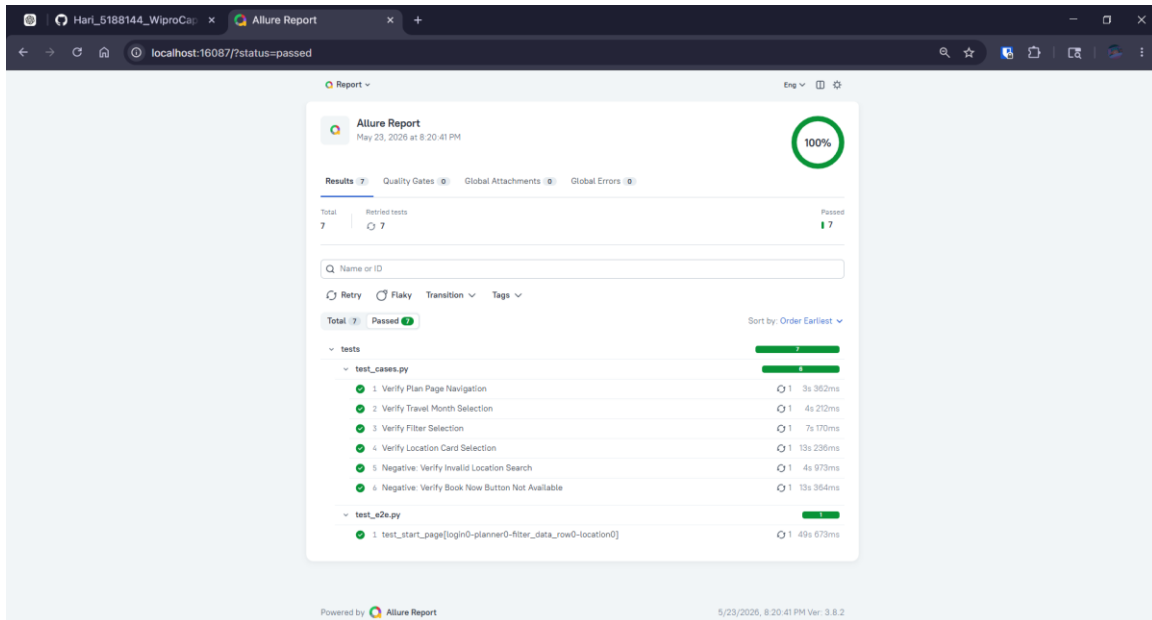
- Enters invalid location value
- Validates application error handling
- Verifies display of:
 - ✚ “Oops! We couldn't find any results for that destination.”
- Ensures unsupported location validation

Negative Test Case 2 - Book Now Button Validation

- Selects unsupported route location
- Navigates to “How To Reach” section
- Verifies “Book Now” button absence
- Ensures unavailable routes cannot be booked

ALLURE REPORTS

- Integrated Allure Reports with PyTest
- Generates professional execution reports
- Displays passed and failed test status
- Captures screenshots during execution
- Maintains execution history
- Displays step-by-step execution flow
- Helps debugging and tracking failures
- Provides detailed execution summary



TEST OUTPUTS

The framework automatically captures screenshots during important execution stages including planner page, filter page, location page, map verification page, and flight details_page.

Detailed execution logs are generated using Python logging framework for debugging and execution tracking.

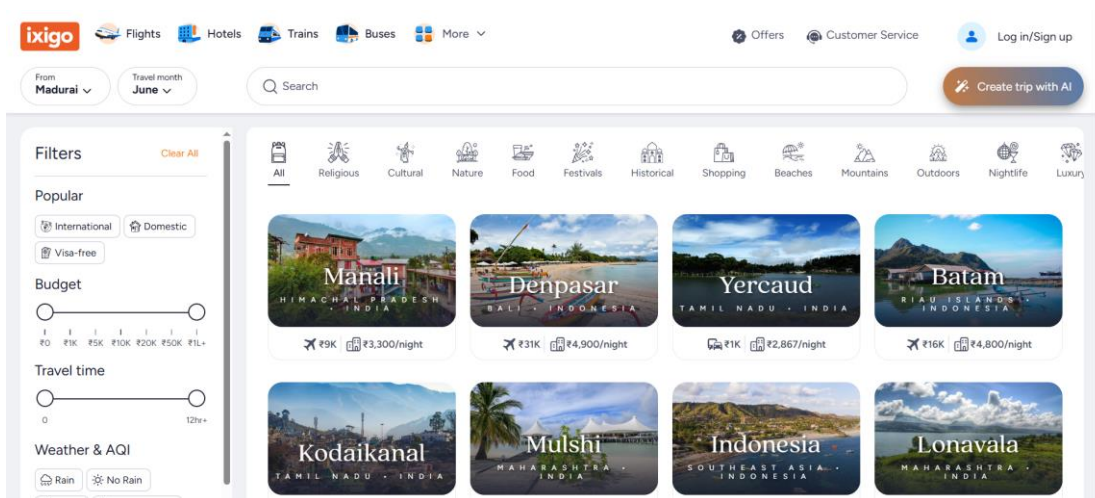


Fig 1: plan_page_navigation

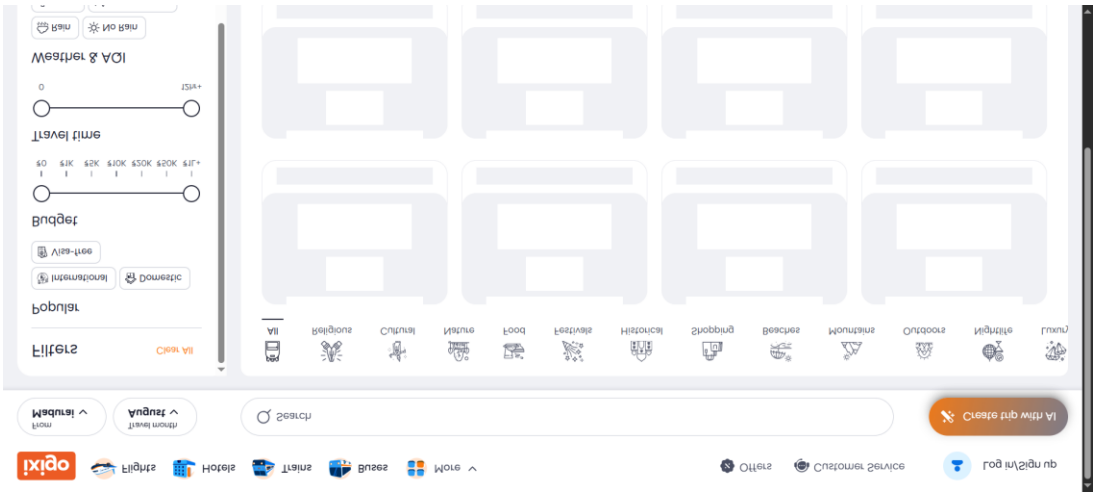


Fig 2: travel_month_selection

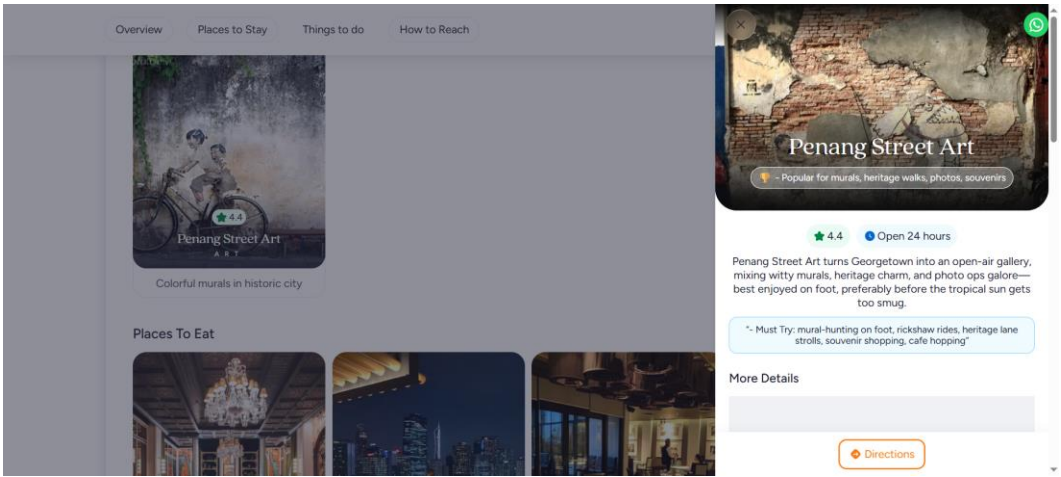


Fig 3: Location card selection

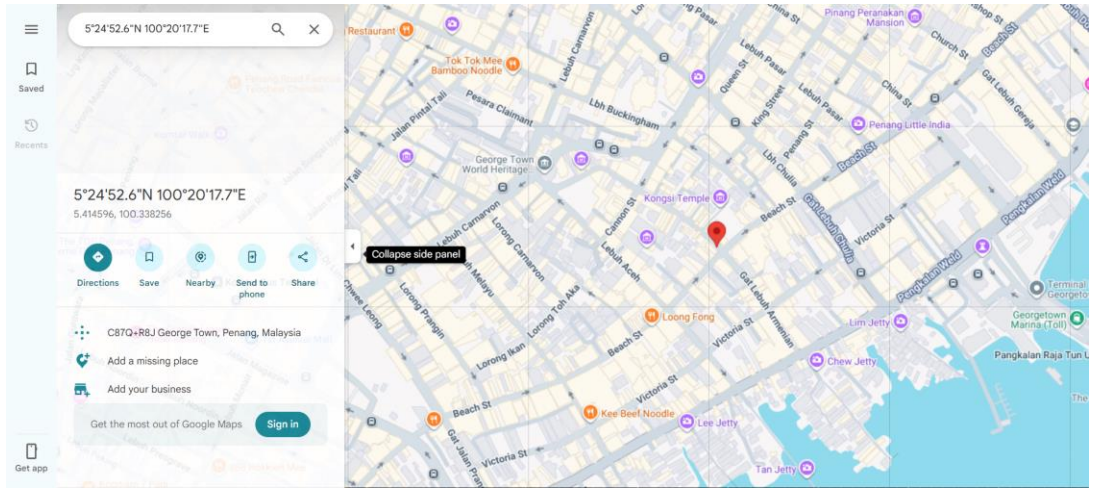


Fig 4: Google Maps page

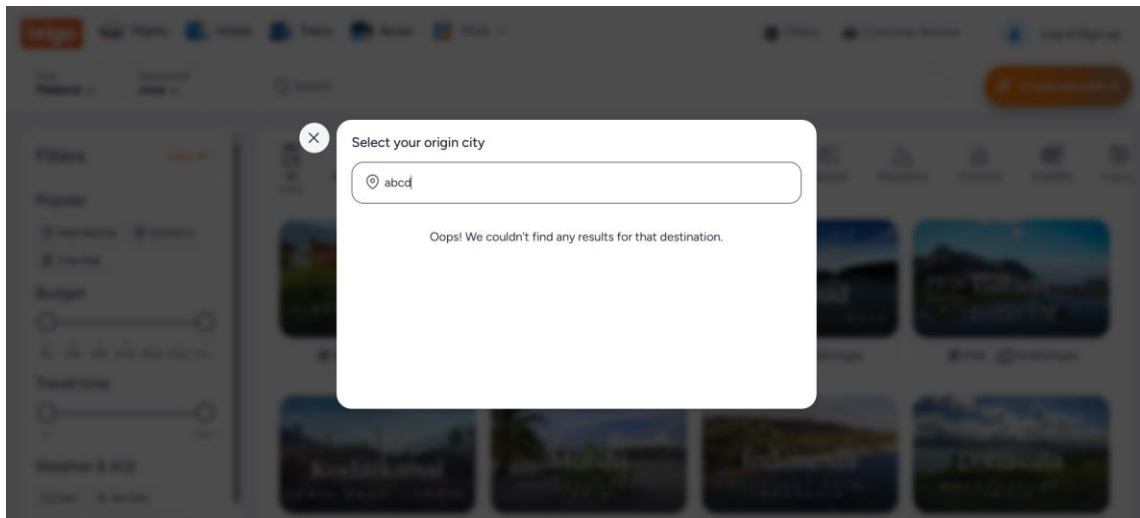


Fig 5: Invalid card selection

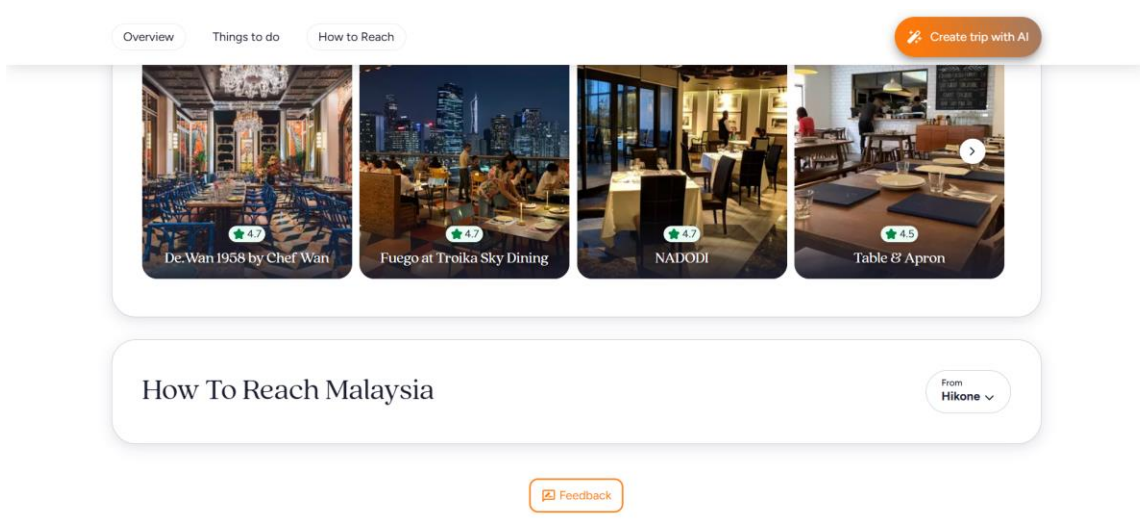


Fig 6: Invalid Book now availability

TEST REPORT

- ✓ Total Tests Executed: 7
- ✓ Tests Passed: 7
- ✓ Tests Failed: 0
- ✓ Automation Accuracy: 100%

CONCLUSION

The Selenium PyTest automation framework successfully validates the ixigo Trip Planner application using End-to-End, Positive and Negative testing approaches.

The framework is designed using:

- Selenium WebDriver
- PyTest Framework
- Page Object Model (POM)
- Allure Reporting
- Logging Framework
- Data-Driven Testing

The automation framework improves:

- Test maintainability
- Reusability
- Scalability
- Reporting quality
- Automation reliability

The framework successfully validates critical business workflows including planner navigation, filter validation, tourist location selection, booking navigation and flight details verification.

All test cases executed successfully with professional reporting, screenshot capture and detailed execution logs.

Overall, the project demonstrates a practical and professional Selenium automation framework for real-world web application testing.

SELENIUM BDD FRAMEWORK

TEST STRATEGY

The project uses Selenium WebDriver with Python Behave BDD framework for automation testing. End-to-End Testing

- Positive Testing
- Negative Testing
- Data-Driven Testing
- Behavior-Driven Development (BDD)

Tools Used:

- Selenium WebDriver
- PyTest
- Allure Reports
- Logging Module

TEST ENVIRONMENT

Operating System	Windows 10
IDE	PyCharm Unified Product
Browser	Google Chrome (or) Edge
Framework	Selenium WebDriver + Behave
Design Pattern	Page Object Model (POM)
Reporting Tool	Allure Reports
Test Approach	Behavior Driven Development

FRAMEWORK ARCHITECTURE

IxigoTripPlanner

```
├─ config
├─ features
│   ├── end_to_end.feature
│   ├── test_cases.feature
│   ├── environment.py
│   └── steps
│       ├── end_to_end.steps
│       └── test_cases.steps
├─ reports
│   ├── allure-results
│   └── allure-report
├─ screenshots
├─ testdata
├─ logs
├─ utils
├─ behave.ini
├─ runtest.bat
└─ requirements.txt
```

TEST EXECUTION

The automation suite includes:

- 1 End-to-End Test Scenario
- 4 Positive Test Scenarios
- 2 Negative Test Scenarios

The End-to-End scenario validates the complete ixigo Trip Planner workflow from planner navigation until flight details verification.

SELENIUM TEST CASE SUMMARY

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- Google Map Verification
- Book Now Navigation
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POSITIVE TEST SCENARIOS

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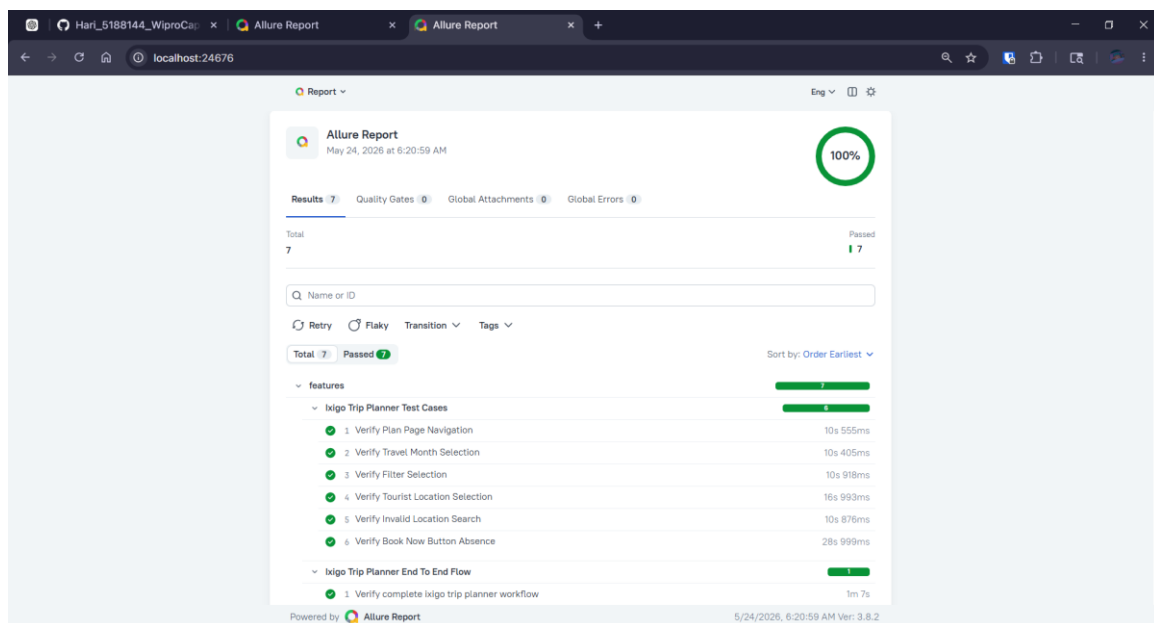
BDD IMPLEMENTATION

The framework follows Behavior Driven Development using:

- Feature Files
- Step Definition Files
- Gherkin Syntax
- Given-When-Then Structure

ALLURE REPORTS

- Integrated Allure Reports with Behave BDD
- Generates professional execution reports
- Displays passed and failed scenario status
- Captures screenshots during execution
- Maintains execution history
- Displays step-by-step scenario execution flow
- Helps debugging and tracking failuresProvides detailed execution summary



TEST OUTPUTS

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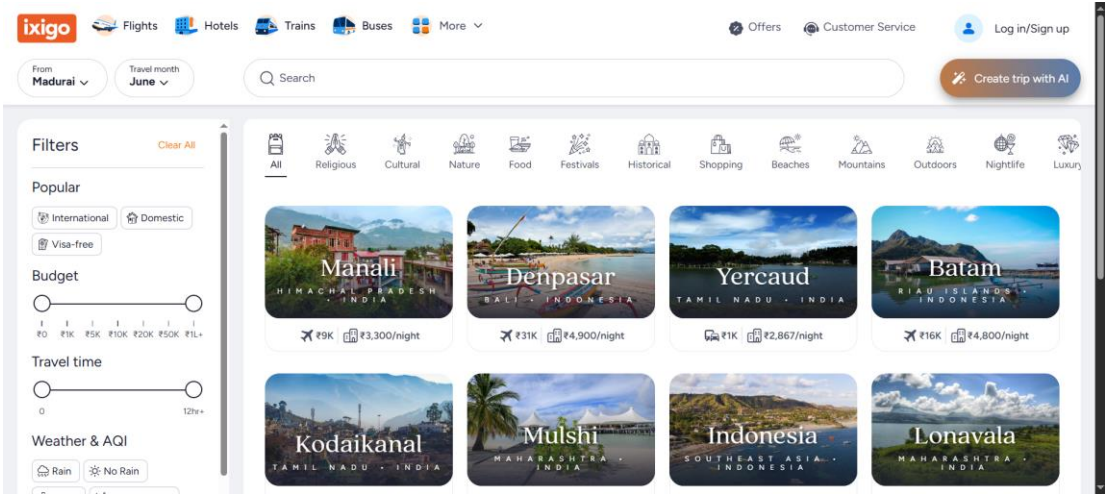


Fig 1: plan_page_navigation

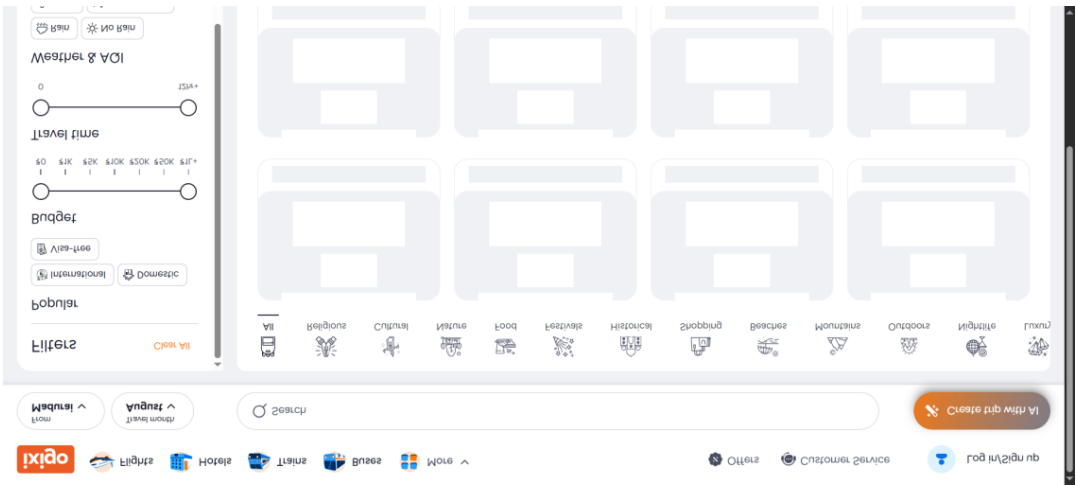


Fig 2: travel_month_selection

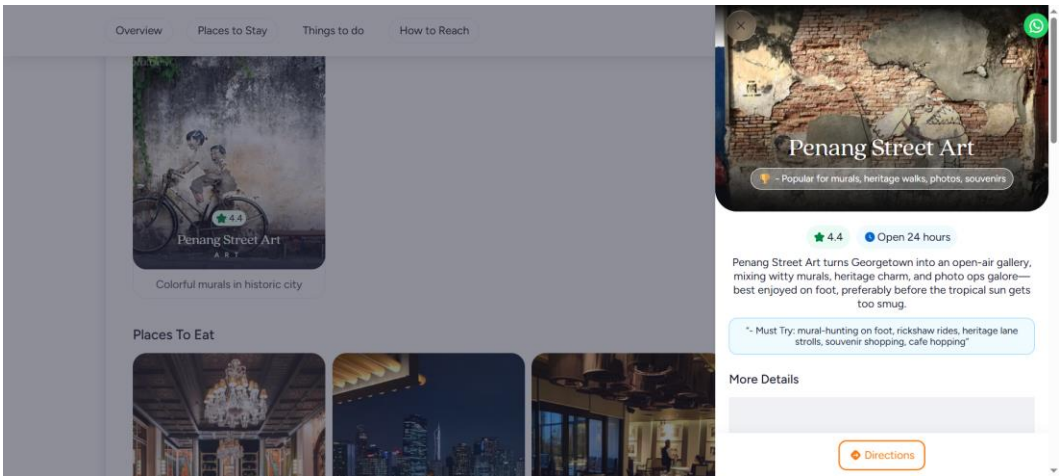


Fig 3: Location card selection

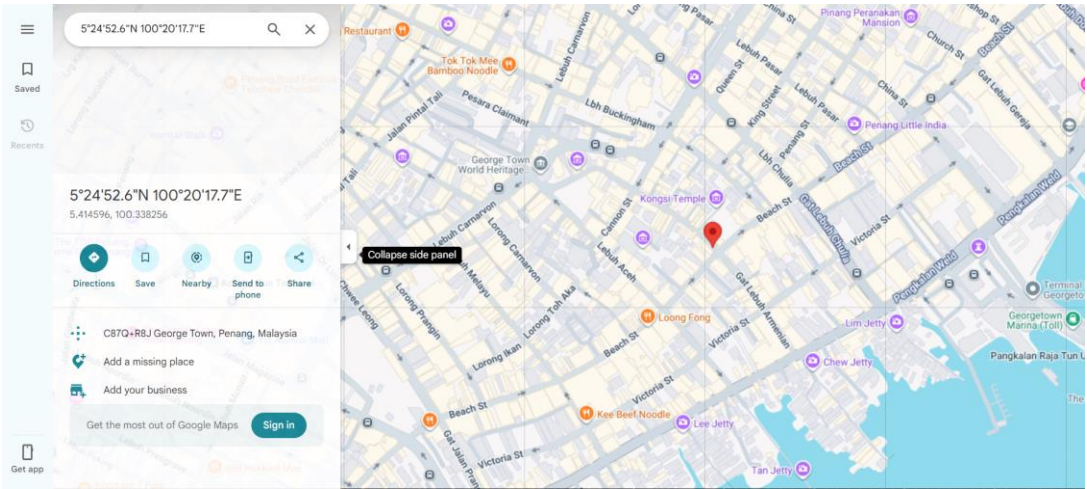


Fig 4: Google Maps page

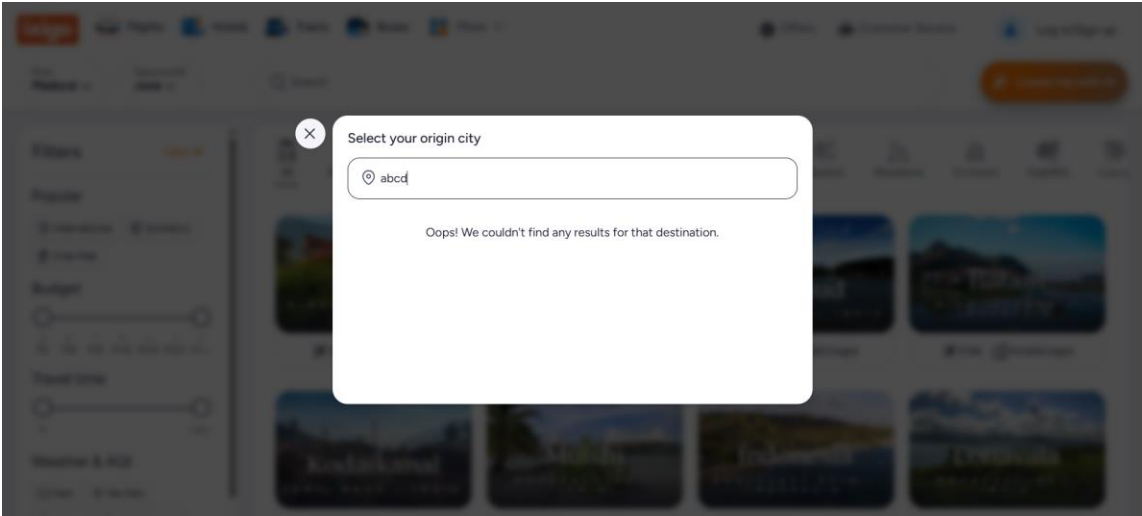


Fig 5: Invalid card selection

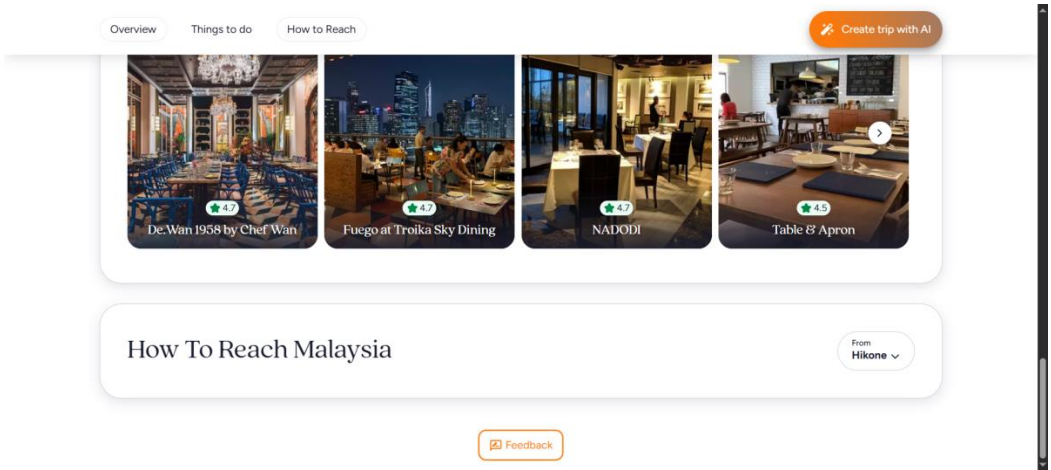


Fig 6: Invalid Book now availability

TEST REPORT

- ✓ Total Scenarios Executed: 7
- ✓ Scenarios Passed: 7
- ✓ Scenarios Failed: 0
- ✓ Automation Accuracy: 100%

CONCLUSION

The Selenium Behave BDD automation framework successfully validates the ixigo Trip Planner application using Behavior Driven Development methodology.

The framework is implemented using:

- Selenium WebDriver
- Behave BDD Framework
- Gherkin Feature Files
- Step Definition Files
- Page Object Model (POM)
- Allure Reporting
- Data-Driven Testing

The framework validates complete business workflows including:

- User Login
- Planner Navigation
- Travel Month Selection
- Filter Application
- Tourist Location Selection
- Google Map Verification
- Booking Navigation
- Flight Details Verification

The implementation of Feature Files and Given-When-Then structure improves readability, maintainability and collaboration between technical and non-technical users.

The framework also includes:

- Professional Allure Reports
- Screenshot Evidence
- Detailed Execution Logs
- Positive and Negative Testing
- Reusable Automation Components

All scenarios executed successfully with accurate validations and detailed reporting.

Overall, the project demonstrates a professional real-world BDD automation framework with scalable architecture, reusable components and reliable automated testing practices.